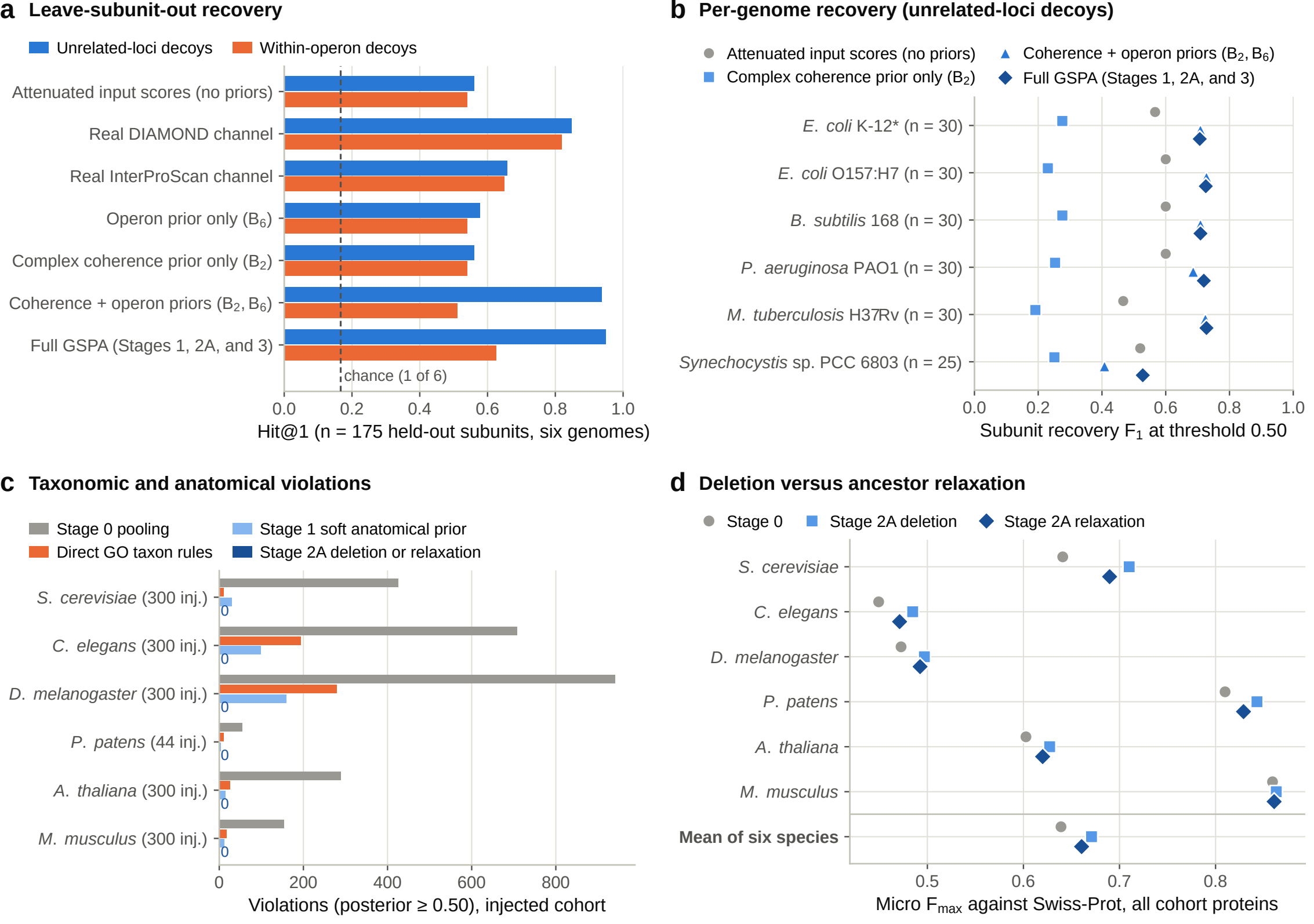
a Leave-subunit-out recovery
Unrelated-loci decoys
Within-operon decoys
Attenuated input scores (no priors)
Real DIAMOND channel
Real InterProScan channel
Operon prior only (B6)
Complex coherence prior only (B2)
Coherence + operon priors (B2, B6)
Full GSPA (Stages 1, 2A, and 3)
chance (1 of 6)
0.0
0.2
0.4
0.6
0.8
1.0
Hit@1 (n = 175 held-out subunits, six genomes)
b Per-genome recovery (unrelated-loci decoys)
Attenuated input scores (no priors)
Complex coherence prior only (B2)
Coherence + operon priors (B2, B6)
Full GSPA (Stages 1, 2A, and 3)
E. coli K-12* (n = 30)
E. coli O157:H7 (n = 30)
B. subtilis 168 (n = 30)
P. aeruginosa PAO1 (n = 30)
M. tuberculosis H37Rv (n = 30)
Synechocystis sp. PCC 6803 (n = 25)
0.0
0.2
0.4
0.6
0.8
1.0
Subunit recovery F1 at threshold 0.50
c Taxonomic and anatomical violations
Stage 0 pooling
Stage 1 soft anatomical prior
Direct GO taxon rules
Stage 2A deletion or relaxation
S. cerevisiae (300 inj.)
C. elegans (300 inj.)
D. melanogaster (300 inj.)
P. patens (44 inj.)
A. thaliana (300 inj.)
M. musculus (300 inj.)
0
0
0
0
0
0
0
200
400
600
800
Violations (posterior ≥ 0.50), injected cohort
d Deletion versus ancestor relaxation
Stage 0
Stage 2A deletion
Stage 2A relaxation
S. cerevisiae
C. elegans
D. melanogaster
P. patens
A. thaliana
M. musculus
Mean of six species
0.5
0.6
0.7
0.8
Micro Fmax against Swiss-Prot, all cohort proteins